

**THE NIGERIAN PHARMACIST, MEDICATION ERRORS AND THE
REGULATION AND CONTROL OF THE PHARMACY PRACTICE BY THE
PHARMACIST'S COUNCIL OF NIGERIA (PCN)**

Being a Text of Presentation

By

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1.0 Introduction

As the concerted battle against the scourge of fake, counterfeit, and sub-standard drugs rages on within the national landscape, the necessity for an integrated and multi-disciplinary approach towards achieving a decisive victory becomes ever more apparent. It constitutes a fundamental challenge for every stakeholder within the health-care profession to ensure that medications and drugs meant for prophylactic, curative, and/or diagnostic purposes within any health-care delivery system are genuine, safe, and efficacious in performing the specific functions for which they were formulated.

A pharmacist is an individual professionally qualified in pharmacy, the branch of health sciences dealing with the preparation, dispensing, and rational use of medicines. This professional is globally recognized and licensed to prepare, compound, mix, and dispense medicines for the ultimate benefit of the patient and the efficacy of the health-care delivery system. Consequently, the role of the Pharmacist in the successful utilization of medicines for therapeutic purposes is total and of paramount importance among all health-care professionals. To put it simply, notwithstanding the availability of the most accurate diagnosis, the most appropriate prescriptions, and indeed the genuineness of the prescribed medicines, so long as the Pharmacist is lacking in professional competence—whether through a dearth of appropriate and relevant updates or a deficit in contemporary professional knowledge—medication errors are most likely to occur.

It is, therefore, incumbent upon the statutory regulatory body responsible for the control and regulation of the practice of pharmacy to ensure that only individuals who are fit, proper, and demonstrably competent are licensed and presented to the public to practice the profession. These pharmaceutical practices traditionally take place within defined premises such as hospital pharmacies, retail community pharmacies, importation/wholesale pharmaceutical establishments, and pharmaceutical manufacturing industries. The Pharmacists Council of Nigeria (PCN) is vested with the authority to register, inspect, approve, and license these pharmaceutical premises where the Pharmacist practices his/her profession. The regulation and control of the practice of the Pharmacy profession in Nigeria by the PCN is exercised on the strength of enabling legislation, as enshrined in the Pharmacists Council of Nigeria Act, Cap. P17, LFN 2004, which was originally promulgated as Decree No. 91 of 1992.

2.0 The Nigerian Pharmacist

The Nigerian pharmacist is defined as the healthcare professional duly registered by the Pharmacists Council of Nigeria (PCN) to practice the pharmacy profession within the country. To achieve this status, an individual must have graduated from any of the presently nine (09) approved Faculties of Pharmacy in Nigerian Universities, undergone the mandatory internship training at a PCN-approved internship centre, and must be annually licensed by the PCN to practice the profession.

Alternatively, a candidate may have been trained overseas in a recognized foreign institution, obtaining at least a Bachelor's degree in Pharmacy, and must subsequently have undertaken and passed the Mandatory Orientation Programme for Foreign-Trained Pharmacists at a centre approved by the PCN. Ensuring the continuous professional competence and fitness to practice of these pharmacists represents a basic and ongoing responsibility of the PCN. The Council achieves this cardinal objective largely through the mounting of Mandatory Continuing Professional Development (MCPD) programmes in its approved provider centres across the nation. The MCPD programme is designed to ensure that pharmacists, as active practitioners, are exposed on a continual basis to current developments and emerging trends in the practice of pharmaceutical sciences. This continuous engagement enables them to maintain improved and up-to-date knowledge and skills, which are essential for effective practice aimed at maximizing benefit for the patient.

The ultimate objective, therefore, has been to present these properly trained, registered, and currently licensed Pharmacists to the Nigerian public for the purpose of dispensing appropriate pharmaceutical products and/or services, as may be required, which can be accessed by members of the public from premises that have themselves been registered, inspected, approved, and licensed by the PCN.

The PCN, on its own part, ensures the appropriate and professional dispensation of these services through a regime of regular monitoring and accreditation of these premises as inspected and approved outlets for pharmacy practice. To ensure strict compliance with its established standards, periodic pharmaceutical inspections are carried out by inspectors duly approved and appointed by the PCN. The purpose of these inspections is to confirm that the set standards by the PCN as the regulatory body are not only adhered to but are also continuously maintained. These inspections could be either announced or

unannounced. Depending on the outcome and recommendations of such inspections, the premises in question could be allowed to continue offering professional pharmaceutical services without hindrance, face temporary restrictions (as in cases of partial accreditation), or be outrightly closed by the application of the Council's official seal. All these regulatory measures are implemented with the primary goal of ensuring that members of the public are protected and have access only to those pharmaceutical premises that have satisfactorily met the minimum standards set by the PCN, as explicitly provided for under the PCN Act 91 of 1992.

3.0 Drug Manufacture, Dispensing and Medication Errors

A medication error is any preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of the health-care professional or the patient. Such events may be related to professional practice, health care products, procedures, and systems, including: prescribing; order communication; product labeling, packaging, and nomenclature; compounding; dispensing; distribution; administration; education; monitoring; and use. The definition of medication errors within the context of this presentation is "pharmacy profession specific," focusing on the responsibilities of the pharmacist within an approved health-care facility.

According to Pharmacists H. K. Abutiate and Edward O. Amporful, both of Ghana, in their paper titled "Medication Errors Prevention: A Pillar for Patient Safety (1)," as published in "The African Pharmacist", Vol. 1, No. 1 of 2008, medication errors are incidents in which there has been an error in the process of prescribing, dispensing, preparing, administering, monitoring, or providing medicine advice, regardless of whether any harm occurred. They stated that an estimated amount of more than £750 million (Seven hundred and fifty million pounds) is lost annually in England from preventable harm caused by medicines. These definitions clearly illustrate that prescription is the inaugural stage of the medication process, and errors originating at this point can propagate problems down the entire drug use chain. The aforementioned authors further noted that the College of Nurses of Ontario's 2006 Medication Practice Standard includes a section on medication errors, which details ways to address the human factors responsible for such errors. This involves rigorously following the five 'rights' - right drug, right dose, right client, right time, and right route.

William A. Zellmer, in his publication, "The Conscience of a Pharmacist - Essays on Vision and Leadership for a Profession", first published in 2002 by the American Society of Health System Pharmacists (ASHP), advocated that "Pharmacists, by virtue of their traditional role in health care, are expected to do what they can to prevent medication errors". With specific reference to hospital Pharmacists, who undoubtedly encounter a higher frequency of medication errors, Zellmer posited that the hospital pharmacist needs to operate on at least three distinct planes in their efforts to reduce medication errors:

- First Plane - Their Own Practice: Learning to recognize the mental lapses that can occur in drug product preparation, distribution, and administration, and building the appropriate checks and controls into their work habits.
- Second Plane - Systemic Design: This deals with hospital systems for ordering, dispensing, and administering medications. Hospital medication systems should be intelligently designed to minimize the potential for human error.
- Third Plane - Quality Assurance: This addresses the need for robust monitoring. He questioned how many pharmacy departments have an ongoing system for monitoring dispensing errors, and how many have a formal process for monitoring medication administration errors?

He conclusively stated that reducing the pervasive problem of medication errors requires unwavering vigilance at many levels within the health care system. Pharmacists should take the lead in developing and implementing multi-disciplinary and assertive programmes specifically designed to prevent medication errors. The more effectively they assume this leadership role, the more lives will be saved.

Undoubtedly, the proper utilization of the unique skills of the Pharmacist—during the formulation of pharmaceutical products (whether in the industry or extemporaneously in the hospital or retail pharmacy) and during the critical counselling of the patient upon filling a prescription or dispensing a product—goes a very long way in addressing and mitigating the incidence of medication errors during each encounter with a patient.

Every product that enters the market from any pharmaceutical manufacturer is accompanied by information. However, the information pertaining to pharmaceutical products is most competently and professionally communicated by the Pharmacist. The guidance on the appropriate use of the medicine by the

patient or user, which is crucial for achieving the intended and desired therapeutic outcome, can only be comprehensively provided by the pharmacist. The act of issuance, as in the dispensing of the pharmaceutical product, takes place in the pharmaceutical premises, such as a retail pharmacy or a hospital pharmacy. These are premises that must be manned by pharmacists who are registered and licensed by the PCN. Furthermore, the premises themselves must have been duly registered, inspected, approved, and licensed by the PCN.

A critical factor towards the reduction of medication errors, therefore, rests squarely with the pharmacists who manufacture and dispense the product. The counselling provided to the patient on the use of a pharmaceutical product must be communicated in a clear, precise, and unambiguous manner to ensure full comprehension and compliance by the patient, thereby achieving the desired therapeutic outcome. Pharmacy practitioners must, as a matter of professional duty, continuously develop themselves by actively participating in the MCPD programmes mounted by the PCN. Default in participating in the appropriate modules of the MCPD, as prescribed by the PCN, leads to the non-licensure of the Pharmacist for practice, notwithstanding their initial registration with the Council.

Similarly, the premises, be it a retail pharmacy and/or hospital pharmacy where the pharmaceutical product is produced and/or dispensed, must have been registered, inspected, approved, and licensed by the PCN. It must also be subject to periodic inspection or re-accreditation to ensure its continuing suitability as a medium or premises appropriately positioned to offer safe and effective pharmaceutical services to the public. The premises must conform to current Good Manufacturing Practice (GMP) to ensure that any likelihood of medication errors, whether through wrong formulation, faulty dispensing, or any other reason, is avoided or reduced to the barest minimum. The premise must be such that only properly formulated pharmaceutical products are manufactured, guaranteed by the availability of adequate facilities for such production. The dispensing of these properly produced pharmaceutical products must be conducted appropriately to the patients and/or end users in such a manner that the pharmacist has effectively conveyed the instructions for the proper use of the medication, leading to compliance through clear and concise communication.

The essential point being emphasized here is that even a properly and ethically produced and well-labelled pharmaceutical product, when dispensed in an inspected and approved pharmaceutical premises, still stands the risk of being

rendered useless and/or ineffective through improper handling or poor dispensing practices by the pharmacist licensed by the regulatory body, the PCN. In other words, the Pharmacist stands squarely as the determinant of the extent to which a dispensed pharmaceutical product is properly used by the patient or end user, which, in turn, directly influences the occurrence or avoidance of medication errors.

This pivotal position of the pharmacist in reducing medication errors is further strongly emphasized in a WHO publication titled "Summary of the Evidence on Patient Safety: Implications for Research" by the World Alliance for Patient Safety (2008). It cited a study conducted in 1999 which demonstrated that when a pharmacist was included on medical wards, medication ordering errors were reduced by a significant 66%, from 10.4 to 3.5 per 1,000 patient days. The tone, language, and gestures employed by the pharmacist in conveying instructions, even for the best of ethically produced drugs, are critical factors that determine whether or not an error will occur with the medication despite its genuineness. Thus, the competence of the pharmacist is critical, and so is the suitability of the premises and the facilities therein.

4.0 Setting Standards for the Practice of Pharmacy by the PCN

The history of pharmacy regulation in Nigeria is long and established. The first pharmacy practice outlet was opened in Nigeria in 1887 at Balogun Street, Lagos. The first legislation affecting the practice of pharmacy or sale of drugs was the Pilotage and Harbor Ordinance of 1878. This was followed by the Hospital Ordinance of 1881 and the Ereko Dispensary Rules of 1889. However, the seminal piece of legislation was the Pharmacy Ordinance No. 8 of 1902. This Ordinance was instrumental to the entry of Mr. Emmanuel Caulcrick into the list of Druggists that same year.

In 1923, the Poison and Pharmacy Ordinance was enacted to control the sale and distribution of drugs. Later, in 1927, the Board of Medical Examiners was established with Dr. A. Robertson as its first Chairman. From 1927 to 1992, the regulatory functions were successively undertaken by the Board of Medical Examiners, the Pharmacy Board of Nigeria, and the Pharmacists Board of Nigeria, each established by a separate legal instrument. The modern era began in 1992 with the promulgation of Decree 91 of 1992 (now an Act of the National Assembly).

The Act establishing the Pharmacists Council of Nigeria (PCN) confers upon the Council the comprehensive powers to control and regulate the practice of pharmacy in all its aspects and ramifications. Section 26 of the PCN Act effectively transfers all the rights and authorization of the repealed law that established the defunct Pharmacists Board of Nigeria (PBN) to the PCN. The Council's authority has been affirmed by the judiciary. For instance, in the case of **Pharmacists Board of Nigeria & 20 Ors Vs. S. O. Adebisin & Co. Ltd (1978)*, the Supreme Court of Nigeria held that the Registrar of PBN is statutorily vested with the power to issue premise licenses. Similarly, the Federal High Court, Lagos in **Suit No. FHC/L/CS/ 1210/02 between Dr. Benedict Okpala & 1 Or Vs. PCN** held that it had not seen the effort of PCN to supervise pharmacists in private and public hospitals as an illegal exercise.

In essence, since 1927, the control and regulation of the practice of pharmacy in Nigeria has been continuously implemented by the various bodies mentioned above. For a period of over one hundred and twenty (120) years, therefore, the PCN and its precursor legislations have consistently defined the parameters for pharmacy practice in Nigeria, whether at the retail, hospital, wholesale/importation, or manufacturing levels.

A landmark event in this regulatory history occurred on Tuesday, August 16, 2005. On this day, the PCN made history when the then Honourable Minister of Health, Prof. Eytayo Lambo, presented to the Nigerian public the first-ever comprehensive publication by the PCN on pharmacy practice standards. The publication, a Four-part Compendium titled ****"Minimum Standards for the Assurance of Pharmaceutical Care"****, represented the first bold and systematic attempt to clearly define the boundaries and scope of pharmacy practice in Nigeria. The compendium articulates detailed minimum standards for the premises where pharmaceutical activities take place, covering Hospital Pharmacy, Retail Pharmacy, Wholesale/Importation, and Pharmaceutical Manufacturing facilities.

The public presentation of this seminal document coincided with the official commissioning of the purpose-built Head Office/Registry of the PCN in Idu, Abuja, by the Honourable Minister who served as the Special Guest of Honour. The historic event was witnessed by a gathering of distinguished dignitaries, including Lady Ufot Ekaette, who chaired the occasion, and representatives from key health agencies such as Prof. Dora N. Akunyili, the Director-General of

NAFDAC, and Dr. Mohammed M. Lecky, then Executive Secretary of the National Health Insurance Scheme (NHIS), among others.

This public presentation firmly positioned the PCN as a regulatory body of repute, one that is proactive and fully abreast of its responsibilities. The Compendium has since been adopted as an indispensable reference material that must be available in any pharmaceutical premises before registration or licensing by the PCN. PCN-approved pharmaceutical inspectors are mandated to verify the presence of at least one copy of the publication during any inspection of a retail pharmacy, hospital pharmacy, pharmaceutical wholesale/importation company, or manufacturing facility. It stands as one of the approved key reference materials that must be present and referred to in any licensed pharmaceutical premises in Nigeria.

5.0 Conclusion

From the foregoing analysis, it is unequivocally clear that the Pharmacist holds a paramount and indispensable role in ensuring patient compliance to medication and, most importantly, in reducing the incidence of medication errors. The team-based approach within the health-care delivery system places the professional responsibility for the production, compounding, and dispensing of pharmaceutical products squarely on the shoulders of the pharmacist. The late great American President, John F. Kennedy, aptly stated: *"The Pharmacist" (implying the pharmacist at the retail outlet) "is an indispensable link in the chain of national health protection. If we did not have the pharmacist, it would be necessary to invent him"* (Pharmacy Practice Management in Developing Communities by Fred B. Adenika, Ph.D, Literamed Publications (Nigeria) Ltd, 1975).

Today, the pharmacist rightly stands tall among health-care professional colleagues in the health-care delivery chain. The role of the PCN is twofold: first, in the training, development, and continuous equipping of the pharmacist with up-to-date knowledge through MCPD; and second, in the regulation of the premises where these services are rendered. Ensuring the availability of a competent, fit, and proper pharmacist, duly registered and annually licensed by the PCN, is a basic statutory responsibility. Similarly, assuring the public that pharmaceutical services are accessed only from premises that have been registered, inspected, approved, and licensed by the PCN is another.

Over the years, the PCN has diligently worked to ensure that the various standards pertaining to the different facets of pharmacy practice under its purview are meticulously documented and disseminated. The availability of these standards ensures that premises offering pharmaceutical services operate within clearly defined boundaries, scope, and benchmarks. The ultimate objective, without any doubt, is to guarantee that the Nigerian public receives the highest possible standard of pharmaceutical care from premises rigorously monitored by the PCN.

Thank you for your attention.

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